

From Schläfli Decomposition to Spectral Bridge: A First Explicit Closure-Projection Channel in V_{600}

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This note introduces no new load-bearing mathematical claim. It synthesises two previously published reproducible exact-arithmetic notes — the Schläfli decomposition of the 600-cell [Smart 2026a] and the 24–600 spectral bridge [Smart 2026b] — and explains their combined role as a first local-to-global spectral channel in the V_{600} programme. Both prior notes are self-contained and reproducible from public repositories; this note is a reader’s map between them.

Abstract

The Schläfli decomposition of the 600-cell represents V_{600} as the binary icosahedral group $2I$ and partitions it into five right cosets of the binary tetrahedral subgroup $2T$, each carrying intrinsic 24-cell structure. The 24–600 spectral bridge then shows that the local $\lambda = 12$ eigenspaces of these five 24-cell sectors zero-extend exactly into the global $\lambda = 12$ eigenspace of the 600-cell Laplacian. This note synthesises those two exact-arithmetic results and explains their combined role as a first explicit local-to-global spectral channel in the V_{600} programme. No new physical claim is made.

1 Why this synthesis note exists

Two reproducible exact-arithmetic notes have been published. Read individually, each is narrow:

- the Schläfli note [Smart 2026a] proves the five-sector finite geometry of the 600-cell;
- the 24–600 spectral bridge [Smart 2026b] proves a single $\lambda = 12$ local-to-global spectral lift.

Read together, they exhibit a pattern: a local shell structure inside the 600-cell does not merely sit inside the global geometry — at a specific spectral value it couples *exactly* to a global invariant sector. The pattern is short and precise enough to warrant a brief synthesis.

This note is a map, not a new claim-heavy paper. Section 2 summarises the Schläfli decomposition; Section 3 summarises the spectral bridge; Section 4 explains why their combination is non-trivial; Section 5 defines the term “closure-projection channel” modestly; Section 6 states what is and is not established; Section 7 explains the role of the synthesis in the broader V_{600} programme; Section 8 lists the dependency map and reproduction status.

2 Artifact I: the five 24-cell cosets

The Schläfli note [Smart 2026a] establishes, by exact $\mathbb{Q}(\sqrt{5})$ arithmetic, that:

$$\begin{aligned} V_{600} &\cong 2I, & 2T &\subset 2I, & [2I : 2T] &= 5, \\ V_{600} &= C_0 \sqcup C_1 \sqcup C_2 \sqcup C_3 \sqcup C_4, & C_i &= 2T \cdot g_i, & |C_i| &= 24. \end{aligned}$$

Each coset C_i is an isometric copy of the 24-cell as a metric subspace of S^3 . Two further structural facts are used below:

- **Two distinct distance shells.** Inside V_{600} , the shortest non-zero squared distance is $d_{600}^2 = (3 - \sqrt{5})/2 \approx 0.382$ (the 600-cell edge); inside each coset C_i , the shortest non-zero squared distance between members is the next shell up, $d_{24}^2 = 1$. The two graphs share *no* edges: no intra-coset pair is at separation d_{600}^2 , and every 600-cell edge runs between two different cosets.
- **Coset action.** Right multiplication by $2I$ permutes the five cosets. The kernel of this action is exactly $\{\pm 1\} \subset 2T$, and the image is $A_5 = 2I/\{\pm 1\}$ acting transitively on five points.

All three facts are certified by exact rational arithmetic; see [Smart 2026a] for the proofs and the reproduction instructions.

3 Artifact II: the $\lambda = 12$ spectral bridge

For each coset C_i let L_{C_i} be the local 24-cell shell Laplacian (the unweighted combinatorial Laplacian of the $d_{24}^2 = 1$ graph on C_i), and let L_{600} be the global 600-cell shell Laplacian (the unweighted combinatorial Laplacian of the d_{600}^2 graph on V_{600}). The bridge note [Smart 2026b] establishes, by exact rational arithmetic, the following *lift identity*:

$$\text{lift}_i(E_{C_i}(12)) \subset E_{600}(12),$$

where $\text{lift}_i: \mathbb{R}^{C_i} \rightarrow \mathbb{R}^{V_{600}}$ is zero-extension and $E_X(\lambda)$ denotes the λ -eigenspace of L_X .

Stacking across the five cosets,

$$E_{\text{lifted}} := \bigoplus_{i=0}^4 \text{lift}_i(E_{C_i}(12)),$$

$$\dim E_{\text{lifted}} = 10, \quad \dim E_{600}(12) = 25.$$

Under the A_5 action of Section 2 (after verifying that $\{\pm 1\}$ acts trivially on $E_{600}(12)$),

$$E_{\text{lifted}} \cong 2 \cdot Y_5, \quad E_{\text{residual}} \cong 3 \cdot Y_5, \quad E_{600}(12) \cong 5 \cdot Y_5,$$

with Y_5 the 5-dimensional irreducible representation of A_5 .

The bridge is also *selective*: $\lambda = 12$ is the only eigenvalue of L_{C_i} that admits a full zero-extension. At $\lambda = 0$ only the global constant (the symmetric sum of the five coset indicators) survives; at $\lambda \in \{4, 8, 10\}$ these are not eigenvalues of L_{600} at all (certified exactly), so the question is vacuous.

4 Why the bridge is non-trivial

Sections 2 and 3 use *different Laplacians on different graphs*. The local Laplacian L_{C_i} is built from the 24-cell edge graph on 24 vertices at squared distance $d_{24}^2 = 1$. The global Laplacian L_{600} is built from the 600-cell edge graph on 120 vertices at squared distance $d_{600}^2 = (3 - \sqrt{5})/2$. These are not subgraphs of each other: no edge of L_{C_i} is an edge of L_{600} , and conversely no L_{600} -edge joins two members of the same coset.

In particular, for a function v supported on a single coset C_i , the value $(L_{600}v)(u)$ at a vertex $u \notin C_i$ is in general non-zero — it is minus the sum of v over the global neighbours of u that happen to lie in C_i . The zero-extension of v is therefore generically *not* a L_{600} -eigenvector.

The content of the lift identity is that, on the local $\lambda = 12$ eigenspace specifically, all such off-coset couplings cancel *identically*: a finite-rank linear condition that turns out to be satisfied at $\lambda = 12$ and nowhere else in the local spectrum.

Put compactly: *the bridge is not a restriction theorem; it is a cancellation theorem*. Equivalently, the $\lambda = 12$ lift is a finite-dimensional compatibility condition between a local shell operator and a global shell operator built from different distance shells.

5 Closure-projection channel: a careful definition

Definition 1 (Closure-projection channel, working). A *closure-projection channel* (in the sense of the present V_{600} programme) is a reproducible local-to-global compatibility relation: a class of modes defined on a local geometric sector projects, lifts, or embeds exactly into a global invariant sector of a larger geometry.

Specialised to the 24–600 case:

The local $\lambda = 12$ eigenspaces on the five 24-cell sectors of the 600-cell zero-extend exactly into the global $\lambda = 12$ eigenspace of L_{600} , and that lift is A_5 -stable.

Definition 1 is *interpretive*. The formal mathematical content is the combination of the two exact results cited in Sections 2 and 3: the five-coset partition and the $\lambda = 12$ lift identity with its A_5 decomposition. The phrase “closure-projection channel” adds no new theorem; it is a name for the pattern.

6 What is and is not established

Established (by the two cited artifacts):

- V_{600} decomposes into five 24-cell cosets (exact).
- The induced coset action descends to A_5 (exact).
- Local 24-cell shell operators and the global 600-cell shell operator are distinct objects on different distance shells (exact).
- The local $\lambda = 12$ eigenspaces zero-extend into the global $\lambda = 12$ sector (exact, $\mathbb{Q}$ -rational).
- The lifted subspace E_{lifted} is A_5 -stable and decomposes as $2 \cdot Y_5$ (exact characters).

Not established (in this note or in the two cited artifacts):

- No derivation of particle masses, the Standard Model, or any specific physical observable.
- No derivation of quantum mechanics or General Relativity.

- No claim that $\lambda = 12$ is physically realised, or that the 600-cell is a physical structure.
- No claim that all closure projection is explained by this example.
- No proof of the Riemann hypothesis or any number-theoretic statement.

The synthesis is mathematical and interpretive only.

7 Why this matters for the V_{600} programme

The V_{600} programme requires explicit mechanisms by which local finite-geometric structures can participate in global invariant sectors. The Schläfli decomposition supplies local sectors; the spectral bridge supplies an exact compatibility channel. Together, they provide a finite test case for the broader idea of *closure projection*.

The next mathematical step is not to identify this channel with a physical observable, but to formalise the general selection principle: *given a local sector, a local operator, a global operator, and a projection/lift map, when does a local eigenspace contribute to a global invariant sector?* The $\lambda = 12$ lift in the 24–600 case is one positive instance and one negative-control template (at every other local eigenvalue) for that general question.

8 Reproducibility and dependency map

This synthesis note introduces no new computation, so it carries no new formal certificate of its own. The dependency on the two prior artifacts is direct:

Artifact	Role	Repro status		
Schläfli decomposition [Smart 2026a]	Proves the five 24-cell cosets and the A_5 coset action	exact	$\mathbb{Q}(\sqrt{5})$ certificate	
24–600 spectral bridge [Smart 2026b]	Proves the $\lambda = 12$ local-to-global lift identity	exact	$\mathbb{Q}$ -rational certificate	
This synthesis note	Explains the combined interpretation	no	new	formal certificate

Both prior artifacts are public:

- <https://github.com/vfd-org/the-24-600-spectral-bridge> — run `pytest closure_transform_engine/tests/` from a clone to re-assert every certificate-level claim of both prior notes.

Although named after the spectral-bridge artifact, this repository now serves as the public V_{600} foundation bundle for all three notes:

- Paper 1 (Schläfli decomposition): `docs/01-schlafl-decomposition/`
- Paper 2 (spectral bridge): `docs/02-spectral-bridge/`
- Paper 3 (this synthesis): `docs/03-closure-projection-channel/`

End of synthesis. The next paper in this programme will attempt to formalise Definition 1 into a general selection principle, but is out of scope here.

References

- [Smart 2026a] L. Smart, *The Schläfli Decomposition of the 600-cell: Five 24-cell Cosets and the Induced A_5 Action*. Public reproducible note, Institute of Vibrational Field Dynamics, 2026. Repository: <https://github.com/vfd-org/the-24-600-spectral-bridge> (folder docs/01-schlafl-li-decomposition/).
- [Smart 2026b] L. Smart, *The 24–600 Spectral Bridge: A selective $\lambda = 12$ eigenspace embedding from five 24-cell cosets into the 600-cell*. Public reproducible note, Institute of Vibrational Field Dynamics, 2026. Repository: <https://github.com/vfd-org/the-24-600-spectral-bridge> (folder docs/02-spectral-bridge/).
- [Coxeter 1973] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973.
- [Conway–Sloane 1999] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer-Verlag, 1999.
- [Conway et al. 1985] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.